

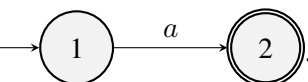
Theories of Computation Formative Assignment 22/23

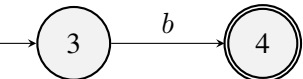
Question 1 Give a regular expression over the alphabet $\Sigma = \{a, b\}$ for the set of words in which the substring 'ab' does not occur. For example, "baa" will be accepted but "aab" will not. [2 marks]

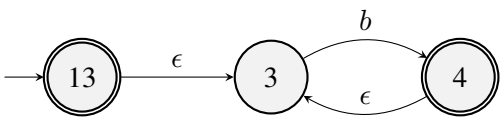
Solution b^*a^*

Question 2 Consider the regular expression $(a|b^*)^*a$. In a step-by-step fashion, using the rules from Handout 2b, convert each sub-expression of the regular expression into an ϵ NFA and, by doing so, obtain a DFA that is equivalent to the overall regular expression. [5 marks]

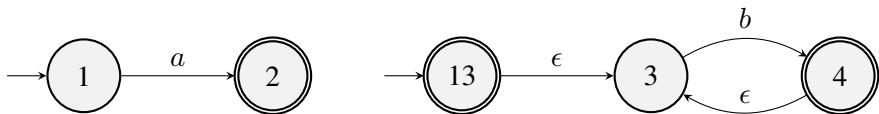
Solution

1. By the single-character rule, a is accepted by .

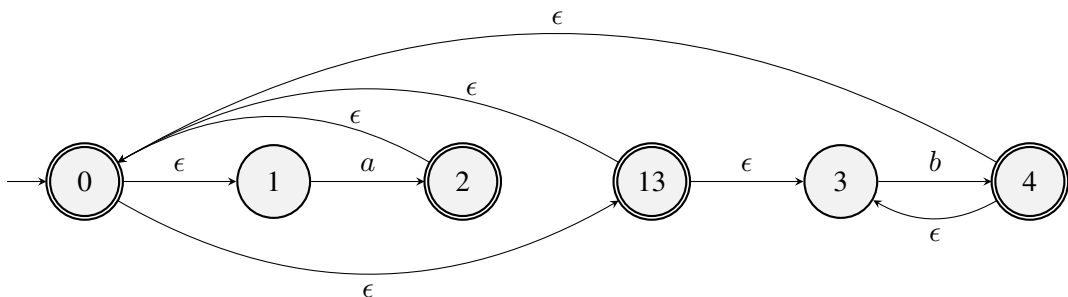
2. By the single-character rule, b is accepted by .

3. Applying the star rule to Step 2, b^* is accepted by .

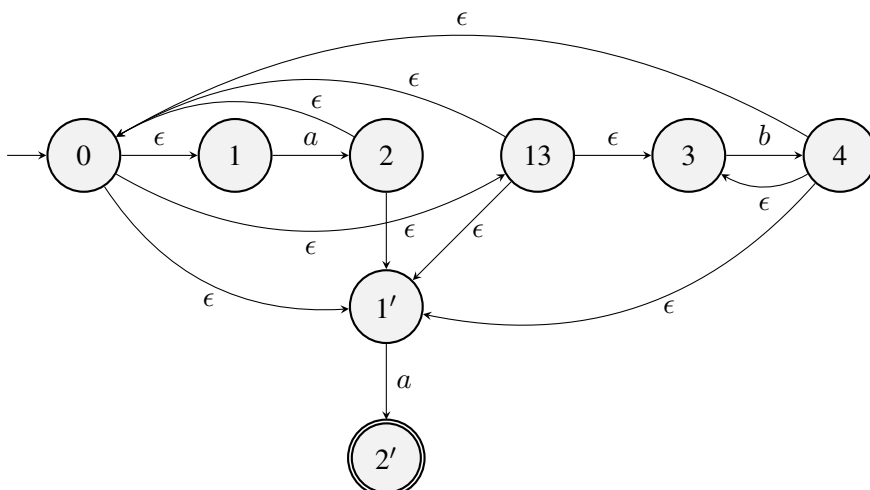
4. Applying the alternative rule to Steps 1 and 3, $(a|b^*)$ is accepted by



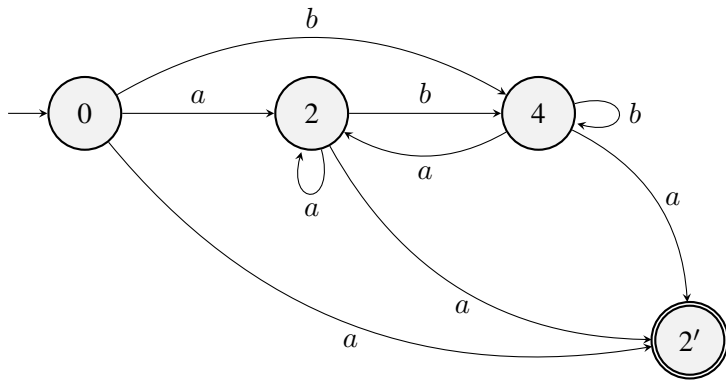
5. Applying the star rule to Step 4, $(a|b^*)^*$ is accepted by



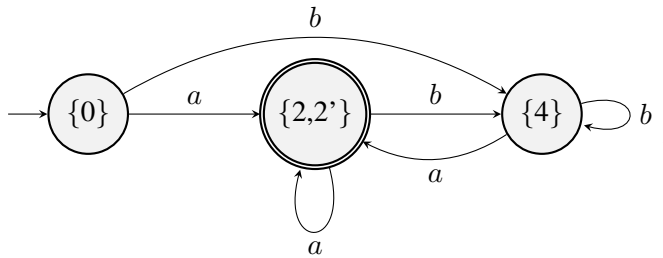
6. Applying the concatenation to Steps 1 and 5, $(a|b^*)^*a$ is accepted by



7. After removing ϵ -transitions, we have



8. After determinising, we have



Question 3 Consider the following sequence of natural numbers $D = (0, 1, 2, 3, 6, 11, 20, \dots)$ defined as $D(0) = 0$, $D(1) = 1$, $D(2) = 2$ and $D(n+3) = D(n) + D(n+1) + D(n+2)$. Show, by induction, that for any natural number n , $D(n) < 2^n$. **[3 marks]**

Solution The base cases are when $n \in \{0, 1, 2\}$. In all three cases, $D(n) < 2^n$ (i.e. $0 < 1$, $1 < 2$, $2 < 4$).

In the inductive case, we want to show that $D(n+3) < 2^{n+3}$; in other words that $D(n) + D(n+1) + D(n+2) < 8(2^n)$. Using the inductive hypothesis, $D(n) < 2^n$, $D(n+1) < 2(2^n)$ and $D(n+2) < 4(2^n)$. Therefore, $D(n) + D(n+1) + D(n+2) < 2^n + 2(2^n) + 4(2^n) = 7(2^n) < 8(2^n)$, and by transitivity we get $D(n) + D(n+1) + D(n+2) < 8(2^n)$, which is what we wanted.